Table of contents

$a_n = 1 + 3n$ 1
$b_n = 1 - (1/2)^n$ 1
$c_n = (-1)^n$ 2
$d_n = (-1)^n/n$ 2

$a_n = 1 + 3n$

Graph of sequence a

The points rise along a straight pattern from (1,4) through (20,61). The graph shows discrete points at the integer indices one through twenty. 2026-10-22T11:08:22.575099 image/svg+xml Matplotlib v3.10.8, https://matplotlib.org/ *{stroke-linejoin: round; stroke-linecap: butt} 1 5 10 15 20 n 0 20 40 60 a n

. . .

The terms grow without bound. The sequence **diverges to** ∞.

$b_n = 1 - (1/2)^n$

Graph of sequence b

The points increase from (1,0.5) toward a dashed horizontal line at one. The graph shows discrete points at the integer indices one through twenty. 2026-10-22T11:08:22.701989 image/svg+xml Matplotlib v3.10.8, https://matplotlib.org/ *{stroke-linejoin: round; stroke-linecap: butt} 1 5 10 15 20 n 0.0 0.5 1.0 b n L = 1

. . .

The terms approach 1. The sequence **converges to** 1.

$c_n = (-1)^n$

Graph of sequence c

Points at odd indices have value minus one; points at even indices have value one. The graph shows discrete points at the integer indices one through twenty. 2026-10-22T11:08:22.804559 image/svg+xml Matplotlib v3.10.8, https://matplotlib.org/ *{stroke-linejoin: round; stroke-linecap: butt} 1 5 10 15 20 n −1 0 1 c n

. . .

The terms alternate between -1 and 1. The sequence **diverges**.

$d_n = (-1)^n/n$

Graph of sequence d

The points alternate below and above zero, with magnitudes decreasing from one to one twentieth. A dashed line marks zero. The graph shows discrete points at the integer indices one through twenty. 2026-10-22T11:08:22.934034 image/svg+xml Matplotlib v3.10.8, https://matplotlib.org/ *{stroke-linejoin: round; stroke-linecap: butt} 1 5 10 15 20 n −1 0 1 d n L = 0

. . .

The signs alternate, but the magnitudes approach 0. The sequence **converges to** 0.